

Math 243: Homework 06

Due Monday, October 5

Problem 1. Find the projection of $\vec{b} = \langle 2, 3 \rangle$ onto $\vec{a} = \langle 1, 4 \rangle$

Problem 2. Compute the distance between the following two points:

$$(5, 8, 15) \quad \text{and} \quad (1, 4, 8)$$

Problem 3.

(a) Choose d so that $A = 2\vec{i} + 3\vec{j}$ is perpendicular to $B = 9\vec{i} + d\vec{j}$

(b) Find a vector C perpendicular to $A = \vec{i} + \vec{j} + \vec{k}$ and $B = \vec{i} - \vec{k}$.

Problem 4. Find a unit vector that has the same direction as the given vector

$$8\vec{i} - \vec{j} + 4\vec{k}$$

Problem 5. Show that the equation

$$x^2 + y^2 + z^2 - 2x - 4y + 8z = 15$$

represents a sphere, and find its center and radius.

Problem 6. Evaluate the following integral:

$$\int_0^{\infty} x e^{-x} dx$$

Problem 7. Find the area enclosed by the curve

$$\begin{cases} r = 2 + \cos(2\theta) \\ 0 \leq \theta \leq 2\pi \end{cases}$$

This curve is shaped like a peanut. See

<https://www.desmos.com/calculator/s2bo1vwb4p>

Problem 8. The magnitude of a velocity vector is called *speed*. A woman walks due west on the deck of a ship at 3 km/h. The ship is moving north at a speed of 22 km/h. Find the speed and direction of the woman relative to the surface of the water.

Problem 9. Evaluate the following integral

$$\int_0^1 (3x + 1)^{\sqrt{2}} dx$$

Problem 10. Consider the triangle whose vertices are following 3 points:

- $A = (4, 6, 8)$
- $B = (1, 2, 3)$
- $C = (5, -1, 3)$

Use vectors to determine whether this triangle is a right triangle.

Problem 11 (Cauchy-Schwarz inequality).

(a) Verify the Cauchy-Schwarz inequality for $\vec{a} = \vec{i} + 2\vec{j} + 2\vec{k}$ and $\vec{b} = 2\vec{i} + 2\vec{j} + \vec{k}$.

(b) What does the inequality become when $\vec{a} = (\sqrt{x}, \sqrt{y})$ and $\vec{b} = (\sqrt{y}, \sqrt{x})$?

Problem 12. Find the projection of $\vec{b} = \langle 12, 1, 2 \rangle$ onto $\vec{a} = \langle -1, 4, 8 \rangle$